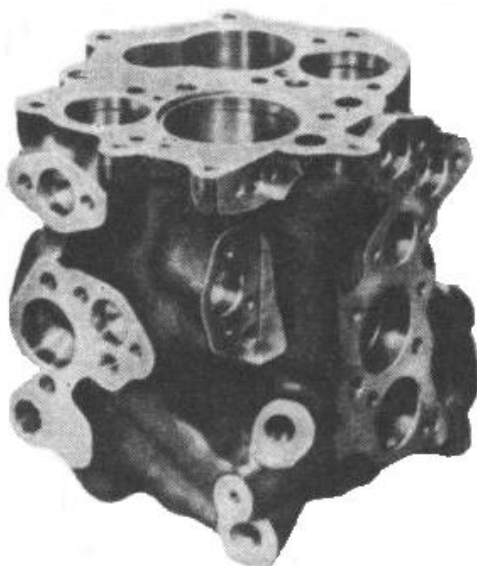


Э.А. ПИЛЬ

**ТЕОРИЯ СЛОЖНОСТИ
ОБРАБОТКИ
КОРПУСНЫХ ДЕТАЛЕЙ
И ЕЁ ПРИМЕНЕНИЕ
В МАШИНО-
И ПРИБОРОСТРОЕНИИ**



**Санкт-Петербург
2003**

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'For me a word of wisdom is more valuable than any gold.'
Democritus



Despite the rapid industrial growth in the early 20th century and the advent of Computer Numerical Control (CNC) machine tools, followed by unmanned machining technology in the form of Flexible Manufacturing Systems (FMS), the issue of part machining complexity has become even more prominent and acute. Any factory process engineer is always faced with the question of the complexity of manufacturing a new part upon receiving its drawing, especially if such parts have not been previously processed in that particular production. Thus, asking any process engineer about the complexity of machining a casing part would likely yield an answer spanning a few sentences. A similar situation exists in the literature on this subject.

Therefore, the time has come to fill this "blank spot" in mechanical processing and provide a clear interpretation of machining complexity by creating a theory that would scientifically explain the complexity of machining any part.

The theory presented in this monograph can also be applied to rotational parts, forging, welding, assembly, and other part classes and types of processing.

This fundamental work is dedicated to my beloved son *Mikhail* and to the memory of my parents and brother.

Respectfully and sincerely yours,